

PHISHING DETECTION DOCUMENTATION

1. Project Name

Phishing URL Detection System

2. Objective

To detect phishing websites by analyzing URLs and protect users from cyber fraud, data theft, and malicious websites.

3. Problem Statement

Phishing attacks are increasing rapidly, and many users unknowingly visit fake websites that steal personal information such as passwords, banking details, and login credentials. Therefore, a system is required to identify phishing URLs before users access them.

4. Research Findings

Research shows that phishing URLs often contain suspicious characters, insecure connections, misleading domain names, and unusual URL structures. These features can be used to detect potentially harmful websites.

5. Methodology

User enters a URL.

The system analyzes the URL.

Security checks are performed.

The URL is classified as Safe or Phishing.

The result is displayed to the user.

6. Tools Used

Python

VS Code

Tkinter

GitHub

7. Implementation

```
url = input("Enter URL: ")
```

```
score = 0
```

```
if len(url) > 50:
```

```
    score += 1
```

```
if "https://" not in url:
```

```
    score += 1
```

```
if "@" in url or "-" in url:
```

```
score += 1
```

```
if score >= 2:  
    print("Phishing URL Detected")  
else:  
    print("Safe URL")
```

8. Screenshots

Home Screen

URL Input Screen

Result Screen

(Insert Screenshots Here)

9. Results

The system successfully identifies suspicious URLs and helps users avoid phishing attacks by providing an alert before visiting harmful websites.

10. Challenges

Detecting newly created phishing websites.

Improving detection accuracy.

Reducing false alerts.

11. Future Scope

Machine Learning Integration

Browser Extension Development

Real-Time URL Monitoring

12. Conclusion

The Phishing URL Detection System is a cybersecurity project that helps users identify malicious websites before sharing sensitive information. It provides a simple and effective solution to reduce phishing attacks and improve online security.